LAB RECORD

1. a) Install Flutter and Dart SDK.

b) Write a simple Dart program to understand the language basics.

Ans:

Pre-Installed

Git ------

1.Open a Web browser

2.visit the official Git website

3. Download the git windows suitable for the operating system.

4.After Download open to setup the git in your system just clicking the next upto to finish.

https://git-scm.com/install/windows

Step 1: Download Flutter SDK

1. Open a web browser.
2. Visit the official Flutter website.
3. Download the Flutter SDK suitable for the operating system.
4. Extract the downloaded Flutter SDK to a suitable location, for example:

Steps:

```
# 1. Download the Flutter SDK
https://flutter.dev/docs/get-started/install

# 2. Extract the zip file to your desired location (e.g., C:\src\flutter).

# 3. Add Flutter to your PATH
# For Windows:
set PATH=%PATH%;C:\src\flutter\bin

# For macOS/Linux:
export PATH="$PATH:/path-to-flutter/bin"
```

The Flutter SDK provides tools and libraries to develop Flutter apps. Make sure the `flutter` command works in your terminal by adding it to the PATH.

Step 2: Set the Environment Variable

Add the Flutter bin directory to the system **PATH**.

C:\src\flutter\bin

Step 3: Verify Flutter Installation

Open **Command Prompt** or **Terminal** and execute:

flutter –version

Step 4: Run Flutter Doctor

Execute the following command:

flutter doctor

Step 5: Install Android Studio

1. Download and install Android Studio.
2. Install the Android SDK and Android SDK Platform Tools.
3. Configure an Android emulator if required.
4. Open Android Studio and install the Flutter and Dart plugins.

3. Set Up Android Studio

Android Studio is required for Flutter development to set up the Android emulator and required tools.

Steps:

```
# 1. Install Android Studio
https://developer.android.com/studio

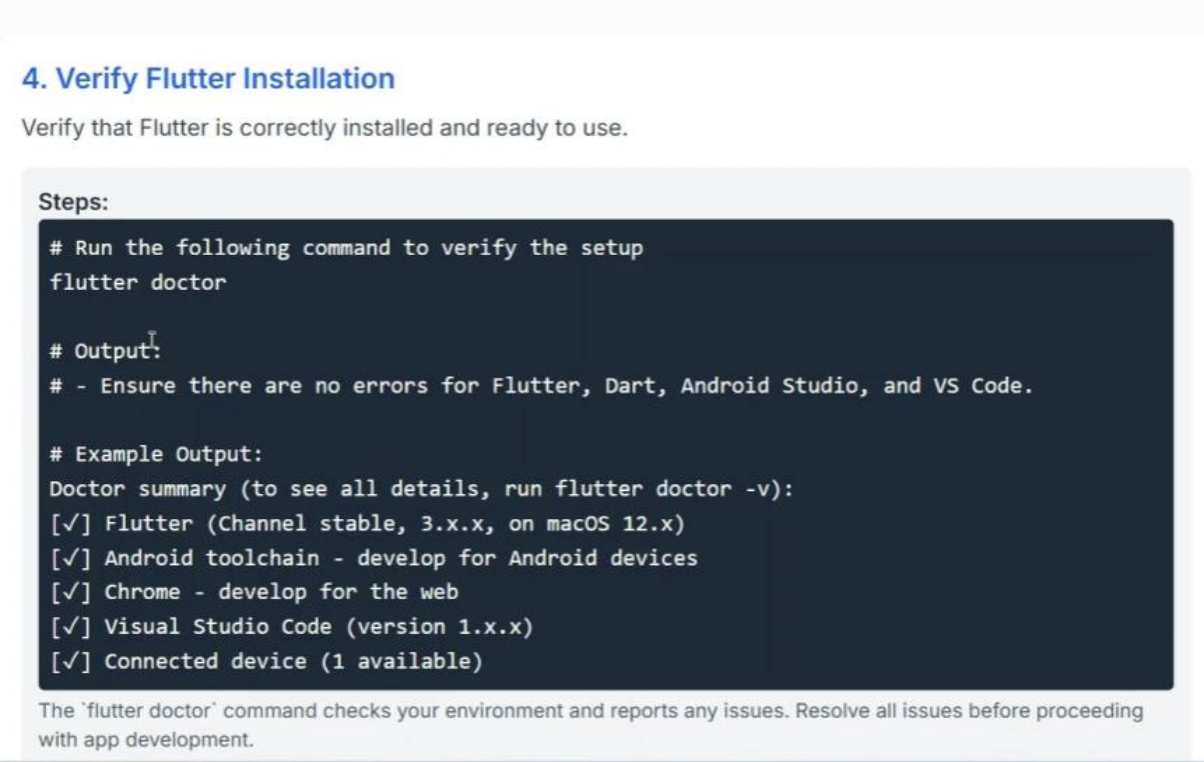
# 2. Install SDK Tools:
# - Open Android Studio
# - Go to Preferences/Settings > Appearance & Behavior > System Settings > Android SDK.
# - Install the 'Android SDK Platform-Tools' and the latest 'Android API'.

# 3. Set Up an Emulator:
# - Go to Tools > AVD Manager.
# - Create a new virtual device with your desired specifications.
```

Android Studio provides an emulator to test your Flutter applications on a virtual Android device.

Step 6: Verify the Development Environment

flutter doctor



installations are completed , now you can open the vs code

Go to the extensions

1.Download Flutter

2.Downlaod Dart

b) Write a simple Dart program to understand the language basics.

1.Open a web browser

2. Visit the Dev pad site to **Run** the sample **dart** programs.

Program:

```
void main() {
  // Variables
  String name = "Ravi";
  int age = 20;
  double marks = 85.5;
  bool passed = true;

  // Display output
```

```
  print("Student Details");
  print("----------------");
  print("Name: $name");
  print("Age: $age");
  print("Marks: $marks");
  print("Passed: $passed");

  // Arithmetic operation
  int nextYearAge = age + 1;
  print("Age next year: $nextYearAge");

  // Conditional statement
  if (marks >= 40) {
    print("Result: PASS");
  } else {
    print("Result: FAIL");
  }
}
```

Now Open Vs Code

1. To create a project

2.Before that you can Create a newfolder in your system(eg:Projects)

3.In Vs Code top search bar just give the greater than symbol(>flutter).

4.You can Application and Select the project folder and name the flutter_application_1 to app.

5.After few seconds ,then can generate the file structure from left side now you open **lib** folder , you can see sample code in main.dart or u can add the newfiles in lib folder only.

6.To Run **the program in Terminal just give the command**

flutter run

2. a) Explore various Flutter widgets (Text, Image, Container, etc.).

Ans:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}


class MyApp extends StatelessWidget {
  const MyApp({super.key});


  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: const Text("Flutter Widgets"),
        ),
```

```
body: SingleChildScrollView(
  padding: const EdgeInsets.all(20),

  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [

      // 1. Text Widget
      const Text(
        "Welcome to Flutter",
        style: TextStyle(
          fontSize: 25,
          fontWeight: FontWeight.bold,
        ),
      ),

      const SizedBox(height: 20),

      // 2. Container Widget
      Container(
        width: double.infinity,
        padding: const EdgeInsets.all(20),
        decoration: BoxDecoration(
          color: Colors.blue,
```

```
    borderRadius: BorderRadius.circular(10),
  ),
  child: const Text(
    "This is a Container",
    style: TextStyle(
      color: Colors.white,
      fontSize: 18,
    ),
  ),
),

const SizedBox(height: 20),

// 3. Image Widget
Image.network(
  "https://picsum.photos/400/200",
  height: 200,
  width: double.infinity,
  fit: BoxFit.cover,
),

const SizedBox(height: 20),

// 4. Icon Widget
const Icon(
```

```
  Icons.favorite,
  color: Colors.red,
  size: 50,
),

const SizedBox(height: 20),

// 5. Card Widget
const Card(
  elevation: 5,
  child: Padding(
    padding: EdgeInsets.all(20),
    child: Text(
      "This is a Card Widget",
      style: TextStyle(fontSize: 18),
    ),
  ),
),

const SizedBox(height: 20),

// 6. Button Widget
ElevatedButton(
  onPressed: () {
    print("Button Clicked");
```

```
            },
            child: const Text("Click Me"),
          ),
        ],
      ),
    ),
  ),
 );
 }
}
```

b) Implement different layout structures using Row, Column, and Stack widgets.

```
import 'package:flutter/material.dart';
void main() {
 runApp(const MyApp());
}

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   debugShowCheckedModeBanner: false,
   home: Scaffold(
    appBar: AppBar(
```

```
  title: const Text("Flutter Layouts"),
  backgroundColor: Colors.blue,
),
body: SingleChildScrollView(
  child: Padding(
    padding: const EdgeInsets.all(20),
    child: Column(
      children: [

        // ROW LAYOUT
        const Text(
          "Row Layout",
          style: TextStyle(fontSize: 24),
        ),

        const SizedBox(height: 15),

        Row(
          mainAxisAlignment:
              MainAxisAlignment.spaceEvenly,
          children: [
            Container(
              width: 80,
              height: 80,
              color: Colors.red,
```

```
    ),
    Container(
      width: 80,
      height: 80,
      color: Colors.green,
    ),
    Container(
      width: 80,
      height: 80,
      color: Colors.blue,
    ),
  ],
),

const SizedBox(height: 30),

// COLUMN LAYOUT
const Text(
  "Column Layout",
  style: TextStyle(fontSize: 24),
),

const SizedBox(height: 15),

Column(
```

```
  children: [
    Container(
      width: 100,
      height: 60,
      color: Colors.orange,
    ),
    const SizedBox(height: 10),
    Container(
      width: 100,
      height: 60,
      color: Colors.purple,
    ),
    const SizedBox(height: 10),
    Container(
      width: 100,
      height: 60,
      color: Colors.teal,
    ),
  ],
),

const SizedBox(height: 30),

// STACK LAYOUT
const Text(
```

```
  "Stack Layout",
  style: TextStyle(fontSize: 24),
),

const SizedBox(height: 15),

SizedBox(
  width: 250,
  height: 200,
  child: Stack(
    children: [
      Container(
        width: 250,
        height: 200,
        color: Colors.yellow,
      ),
      Positioned(
        top: 30,
        left: 40,
        child: Container(
          width: 170,
          height: 130,
          color: Colors.blue,
        ),
      ),
```

```
                Positioned(
                  top: 65,
                  left: 80,
                  child: Container(
                    width: 90,
                    height: 60,
                    color: Colors.red,
                  ),
                ),
              ],
            ),
          ),
        ],
      ),
    ),
  ),
 );
 }
}
```

3. a) Design a responsive UI that adapts to different screen sizes.

```
import 'package:flutter/material.dart';
void main() {
```

```
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {

    // Get screen width
    double width = MediaQuery.of(context).size.width;

    return Scaffold(
```

```
    appBar: AppBar(
      title: const Text("Responsive UI"),
    ),

    body: Center(
      child: width < 600
          ? Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                box("Home"),
                box("Profile"),
                box("Settings"),
              ],
            )
          : Row(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                box("Home"),
                box("Profile"),
                box("Settings"),
              ],
            ),
    ),
  );
}
```

```
  // Reusable widget
  Widget box(String title) {
    return Container(
      width: 120,
      height: 80,
      margin: const EdgeInsets.all(10),
      alignment: Alignment.center,
      color: Colors.blue,
      child: Text(
        title,
        style: const TextStyle(
          color: Colors.white,
          fontSize: 18,
        ),
      ),
    );
  }
}
```

Errors u can get to modify the lengths

Mobile → 390 pixels

Tablet → 800 pixels

Desktop → 1200 pixels

b) Implement media queries and breakpoints for responsiveness.

```
import 'package:flutter/material.dart';
```

```
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  // Create a reusable box
  Widget itemBox(String name) {
    return Container(
      height: 100,
      margin: const EdgeInsets.all(8),
      alignment: Alignment.center,
```

```
    decoration: BoxDecoration(
      color: Colors.blue,
      borderRadius: BorderRadius.circular(10),
    ),
    child: Text(
      name,
      style: const TextStyle(
        color: Colors.white,
        fontSize: 18,
      ),
    ),
  );
}

@override
Widget build(BuildContext context) {

  // Get screen width
  double width = MediaQuery.of(context).size.width;

  int columns;

  // Breakpoints
  if (width < 600) {
    columns = 1;       // Mobile
```

```
} else if (width < 1000) {
  columns = 2;       // Tablet
} else {
  columns = 3;       // Desktop
}

return Scaffold(
  appBar: AppBar(
    title: const Text("Media Query Demo"),
  ),

  body: Padding(
    padding: const EdgeInsets.all(10),

    child: Column(
      children: [

        Text(
          "Screen Width: ${width.toStringAsFixed(0)}",
          style: const TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ),
        ),
```

```
            const SizedBox(height: 20),

            Expanded(
              child: GridView.count(
                crossAxisCount: columns,
                children: [
                  itemBox("Home"),
                  itemBox("Profile"),
                  itemBox("Courses"),
                  itemBox("Students"),
                  itemBox("Faculty"),
                  itemBox("Settings"),
                ],
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

4. a) Set up navigation between different screens using Navigator.

```
import 'package:flutter/material.dart';
void main() {
```

```
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomeScreen(),
    );
  }
}

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Home Screen"),
      ),
```

```
      body: Center(
        child: ElevatedButton(
          child: const Text("Go to Second Screen"),

          onPressed: () {

            // Navigate to Second Screen
            Navigator.push(
              context,
              MaterialPageRoute(
                builder: (context) => const SecondScreen(),
              ),
            );
          },
        ),
      ),
    );
  }
}

// ---------------- SECOND SCREEN ----------------

class SecondScreen extends StatelessWidget {
  const SecondScreen({super.key});
```

```
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Second Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go Back"),

          onPressed: () {

            // Return to previous screen
            Navigator.pop(context);
          },
        ),
      ),
    );
  }
}
```

b) Implement navigation with named routes.

```
import 'package:flutter/material.dart';
```

```
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,

      // Define named routes
      initialRoute: '/',

      routes: {
        '/': (context) => const HomeScreen(),
        '/profile': (context) => const ProfileScreen(),
        '/settings': (context) => const SettingsScreen(),
      },
    );
  }
}

// ---------- HOME SCREEN ----------
```

```
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Home Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go to Profile"),

          onPressed: () {
            Navigator.pushNamed(
              context,
              '/profile',
            );
          },
        ),
      ),
    );
  }
```

```
}

// ---------- PROFILE SCREEN ----------

class ProfileScreen extends StatelessWidget {
  const ProfileScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Profile Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go to Settings"),

          onPressed: () {
            Navigator.pushNamed(
              context,
              '/settings',
            );
          },
        ),
```

```
      ),
    );
  }
}

// ---------- SETTINGS SCREEN ----------

class SettingsScreen extends StatelessWidget {
  const SettingsScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Settings Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go Back"),

          onPressed: () {
            Navigator.pop(context);
          },
        ),
```

```
    ),
   );
  }
}
```

5. a) Learn about stateful and stateless widgets.

What is a StatelessWidget?

A StatelessWidget is a widget whose data does not change while the application is running.

```
class Welcome extends StatelessWidget {
  const Welcome({super.key});
  @override
  Widget build(BuildContext context) {
    return const Text("Welcome to Flutter");
  }
}
```

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

// ---------- STATELESS WIDGET ----------

class MyApp extends StatelessWidget {
  const MyApp({super.key});
```

```
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const CounterPage(),
    );
  }
}
// ---------------- STATEFUL WIDGET ----------------
class CounterPage extends StatefulWidget {
  const CounterPage({super.key});
  @override
  State<CounterPage> createState() => _CounterPageState();
}
// State of CounterPage
class _CounterPageState extends State<CounterPage> {

  int count = 0;

  // Increase counter
  void increase() {
    setState(() {
      count++;
    });
  }
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("Stateful Widget"),
    ),

    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          // Stateless Text
          const Text(
            "Counter Value",
            style: TextStyle(
              fontSize: 25,
              fontWeight: FontWeight.bold,
            ),
          ),

          const SizedBox(height: 20),

          // Changing value
```

```
            Text(
              "$count",
              style: const TextStyle(
                fontSize: 50,
                color: Colors.blue,
              ),
            ),

            const SizedBox(height: 20),

            ElevatedButton(
              onPressed: increase,
              child: const Text("Increase"),
            ),
          ],
        ),
      ),
    );
  }
}
```

b) Implement state management using set State and Provider.

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
```

```
void main() {
  runApp(
    ChangeNotifierProvider(
      create: (context) => CounterProvider(),
      child: const MyApp(),
    ),
  );
}

// ---------------- PROVIDER ----------------

class CounterProvider extends ChangeNotifier {

  int count = 0;

  void increase() {
    count++;
    notifyListeners();
  }

  void decrease() {
    count--;
    notifyListeners();
  }
```

```
}

// ---------------- APP ----------------

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const CounterPage(),
    );
  }
}

// ---------------- UI ----------------

class CounterPage extends StatelessWidget {
  const CounterPage({super.key});

  @override
  Widget build(BuildContext context) {

    final counter = Provider.of<CounterProvider>(context);
```

```
return Scaffold(
  appBar: AppBar(
    title: const Text("Provider Example"),
  ),

  body: Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [

        const Text(
          "Counter",
          style: TextStyle(
            fontSize: 25,
          ),
        ),

        const SizedBox(height: 20),

        Text(
          "${counter.count}",
          style: const TextStyle(
            fontSize: 50,
            fontWeight: FontWeight.bold,
```

```
        ),
      ),

      const SizedBox(height: 20),

      Row(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          ElevatedButton(
            onPressed: counter.decrease,
            child: const Text("-"),
          ),

          const SizedBox(width: 20),

          ElevatedButton(
            onPressed: counter.increase,
            child: const Text("+"),
          ),
        ],
      ),
    ],
  ),
),
```

```
    );
  }
}
```

Firebase Setup with Flutter

1.Create a Firebase project and connect it with a Flutter application.

Ans:

1. Open a web browser
2. https://firebase.google.com
3. Create a project in Firebase Console.
4. Create a Flutter application.
5. Configure Firebase for Android.
6. Add the required Firebase packages.
7. Initialize Firebase in main.dart.
8. Run the application and verify the Firebase connection.
9. Must required the node js in the system and to set path , before linking the project to check the version in cmd.

    node –v

    npm –v

    flutter –version

    flutter doctor

2. **Configure Firebase in an existing Flutter project**

Open cmd , before running the commands in cmd ,vs code terminal once

check the project path.

firebase login

dart pub global activate flutterfire_cli

flutterfire configure

account was login successful it shows your gmail.

Then check the projects in the firebase

firebase projects:list

3. Create a Flutter application and integrate Firebase Core.

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'firebase_options.dart';
void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );

  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: const Text("Firebase Demo"),
        ),
        body: const Center(
          child: Text("Firebase Connected"),
        ),
```

```
      ),
    );
  }
}
```

U can get package error just use the command in terminal.

dart pub add firebase_core

then run the application,

flutter run

4.Implement user registration using Firebase Email and Password Authentication.

- Name field
- Email field
- Password field
- Register button

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'firebase_options.dart';
void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
```

```
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Firebase Registration',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const RegisterPage(),
    );
  }
}

class RegisterPage extends StatefulWidget {
  const RegisterPage({super.key});

  @override
  State<RegisterPage> createState() => _RegisterPageState();
}

class _RegisterPageState extends State<RegisterPage> {
  final TextEditingController nameController = TextEditingController();
  final TextEditingController emailController = TextEditingController();
  final TextEditingController passwordController = TextEditingController();

  bool isLoading = false;

  Future<void> registerUser() async {
    String name = nameController.text.trim();
    String email = emailController.text.trim();
    String password = passwordController.text.trim();
```

```
if (name.isEmpty || email.isEmpty || password.isEmpty) {
  ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
      content: Text('Please fill all fields'),
    ),
  );
  return;
}

if (password.length < 6) {
  ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
      content: Text('Password must contain at least 6 characters'),
    ),
  );
  return;
}

setState(() {
  isLoading = true;
});

try {
  UserCredential userCredential =
      await FirebaseAuth.instance.createUserWithEmailAndPassword(
    email: email,
    password: password,
  );

  print('User ID: ${userCredential.user!.uid}');
```

```
  ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
      content: Text('Registration successful!'),
    ),
  );

  nameController.clear();
  emailController.clear();
  passwordController.clear();
} on FirebaseAuthException catch (e) {
  String message = 'Registration failed';

  if (e.code == 'email-already-in-use') {
    message = 'This email is already registered.';
  } else if (e.code == 'invalid-email') {
    message = 'Please enter a valid email address.';
  } else if (e.code == 'weak-password') {
    message = 'Password is too weak.';
  }

  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(
      content: Text(message),
    ),
  );
} catch (e) {
  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(
      content: Text('Error: $e'),
```

```
      ),
    );
  } finally {
    setState(() {
      isLoading = false;
    });
  }
}

@override
void dispose() {
  nameController.dispose();
  emailController.dispose();
  passwordController.dispose();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('User Registration'),
      centerTitle: true,
    ),

    body: Padding(
      padding: const EdgeInsets.all(20),

      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
```

```
children: [

  const Text(
    'Create Account',
    style: TextStyle(
      fontSize: 28,
      fontWeight: FontWeight.bold,
    ),
  ),

  const SizedBox(height: 30),

  // Name
  TextField(
    controller: nameController,
    decoration: const InputDecoration(
      labelText: 'Name',
      prefixIcon: Icon(Icons.person),
      border: OutlineInputBorder(),
    ),
  ),

  const SizedBox(height: 15),

  // Email
  TextField(
    controller: emailController,
    keyboardType: TextInputType.emailAddress,
    decoration: const InputDecoration(
      labelText: 'Email',
```

```
    prefixIcon: Icon(Icons.email),
    border: OutlineInputBorder(),
  ),
),

const SizedBox(height: 15),

// Password
TextField(
  controller: passwordController,
  obscureText: true,
  decoration: const InputDecoration(
    labelText: 'Password',
    prefixIcon: Icon(Icons.lock),
    border: OutlineInputBorder(),
  ),
),

const SizedBox(height: 25),

// Register Button
SizedBox(
  width: double.infinity,
  height: 50,
  child: ElevatedButton(
    onPressed: isLoading ? null : registerUser,
    child: isLoading
        ? const CircularProgressIndicator(
            color: Colors.white,
          )
```

```
                : const Text(
                    'Register',
                    style: TextStyle(fontSize: 18),
                  ),
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

U can get package error just use the command in terminal.

dart pub add firebase_auth

then run the application,

flutter run

5. Create a Firestore collection and add student details from a Flutter application.

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

import 'firebase_options.dart';
void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );
```

```
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Student Management',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const StudentPage(),
    );
  }
}

class StudentPage extends StatefulWidget {
  const StudentPage({super.key});

  @override
  State<StudentPage> createState() => _StudentPageState();
}

class _StudentPageState extends State<StudentPage> {
  final TextEditingController nameController = TextEditingController();
  final TextEditingController emailController = TextEditingController();
  final TextEditingController phoneController = TextEditingController();
```

```
final TextEditingController courseController = TextEditingController();

Future<void> addStudent() async {
  String name = nameController.text.trim();
  String email = emailController.text.trim();
  String phone = phoneController.text.trim();
  String course = courseController.text.trim();

  if (name.isEmpty ||
      email.isEmpty ||
      phone.isEmpty ||
      course.isEmpty) {
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(
        content: Text('Please fill all fields'),
      ),
    );
    return;
  }

  try {
    await FirebaseFirestore.instance.collection('students').add({
      'name': name,
      'email': email,
      'phone': phone,
      'course': course,
      'createdAt': FieldValue.serverTimestamp(),
    });

    ScaffoldMessenger.of(context).showSnackBar(
```

```
      const SnackBar(
        content: Text('Student added successfully'),
      ),
    );

    nameController.clear();
    emailController.clear();
    phoneController.clear();
    courseController.clear();
  } catch (e) {
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(
        content: Text('Error: $e'),
      ),
    );
  }
}

@override
void dispose() {
  nameController.dispose();
  emailController.dispose();
  phoneController.dispose();
  courseController.dispose();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
```

```
appBar: AppBar(
  title: const Text('Student Details'),
  centerTitle: true,
),

body: Padding(
  padding: const EdgeInsets.all(20),

  child: Column(
    children: [

      TextField(
        controller: nameController,
        decoration: const InputDecoration(
          labelText: 'Student Name',
          prefixIcon: Icon(Icons.person),
          border: OutlineInputBorder(),
        ),
      ),

      const SizedBox(height: 15),

      TextField(
        controller: emailController,
        keyboardType: TextInputType.emailAddress,
        decoration: const InputDecoration(
          labelText: 'Email',
          prefixIcon: Icon(Icons.email),
          border: OutlineInputBorder(),
        ),
```

```
),

const SizedBox(height: 15),

TextField(
  controller: phoneController,
  keyboardType: TextInputType.phone,
  decoration: const InputDecoration(
    labelText: 'Phone',
    prefixIcon: Icon(Icons.phone),
    border: OutlineInputBorder(),
  ),
),

const SizedBox(height: 15),

TextField(
  controller: courseController,
  decoration: const InputDecoration(
    labelText: 'Course',
    prefixIcon: Icon(Icons.school),
    border: OutlineInputBorder(),
  ),
),

const SizedBox(height: 25),

SizedBox(
  width: double.infinity,
  height: 50,
```

```
            child: ElevatedButton(
              onPressed: addStudent,

              child: const Text(
                'Add Student',
                style: TextStyle(fontSize: 18),
              ),
            ),
          ),
        ],
      ),
    ),
  );
 }
}
```

U can get package error just use the command in terminal.

dart pub add cloud_storage

then run the application,

flutter run